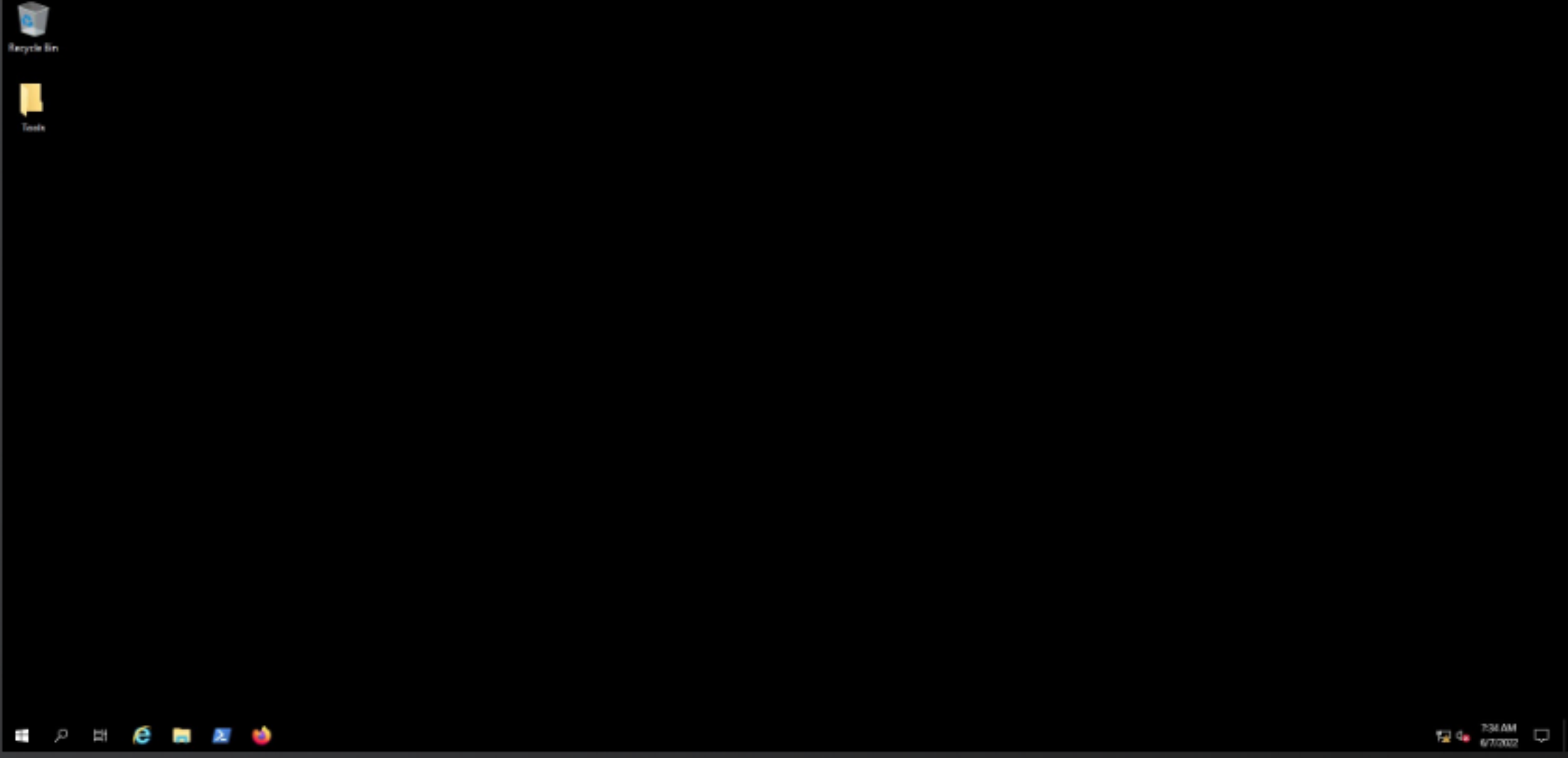


SOLUTIONS



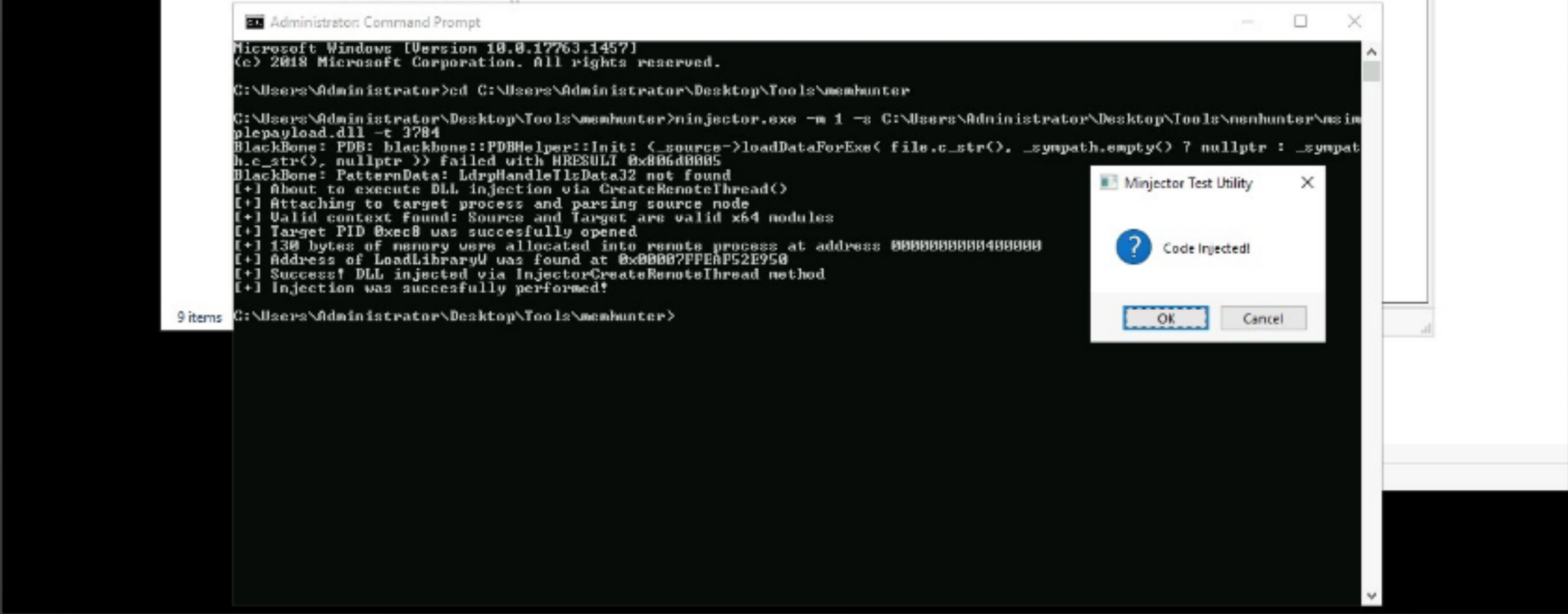
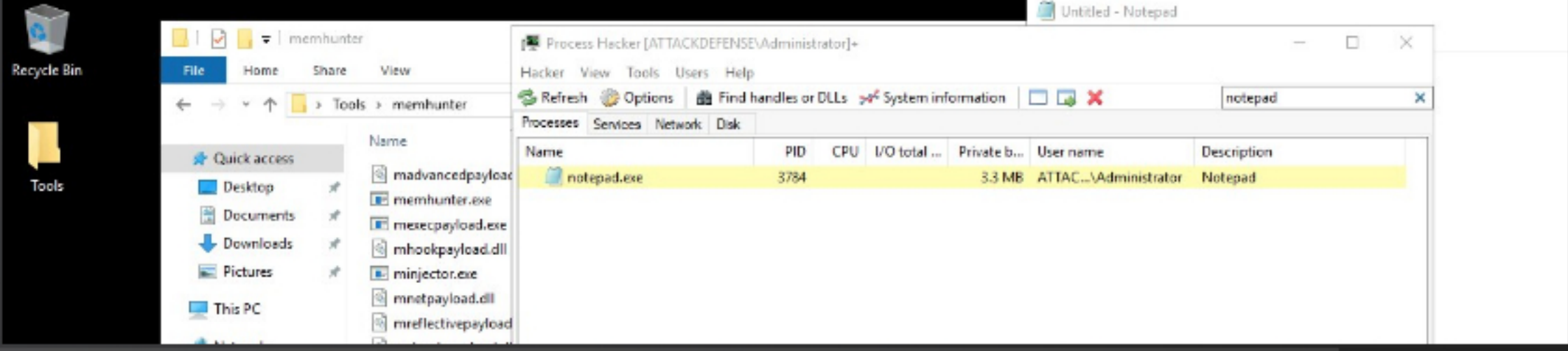
Below, you can find solutions for each task. Remember though, that you can follow your own strategy, which may be different from the one explained in the following lab.

Task 1. Detect detecting process injection using memhunter, minjector & Process Hacker 2

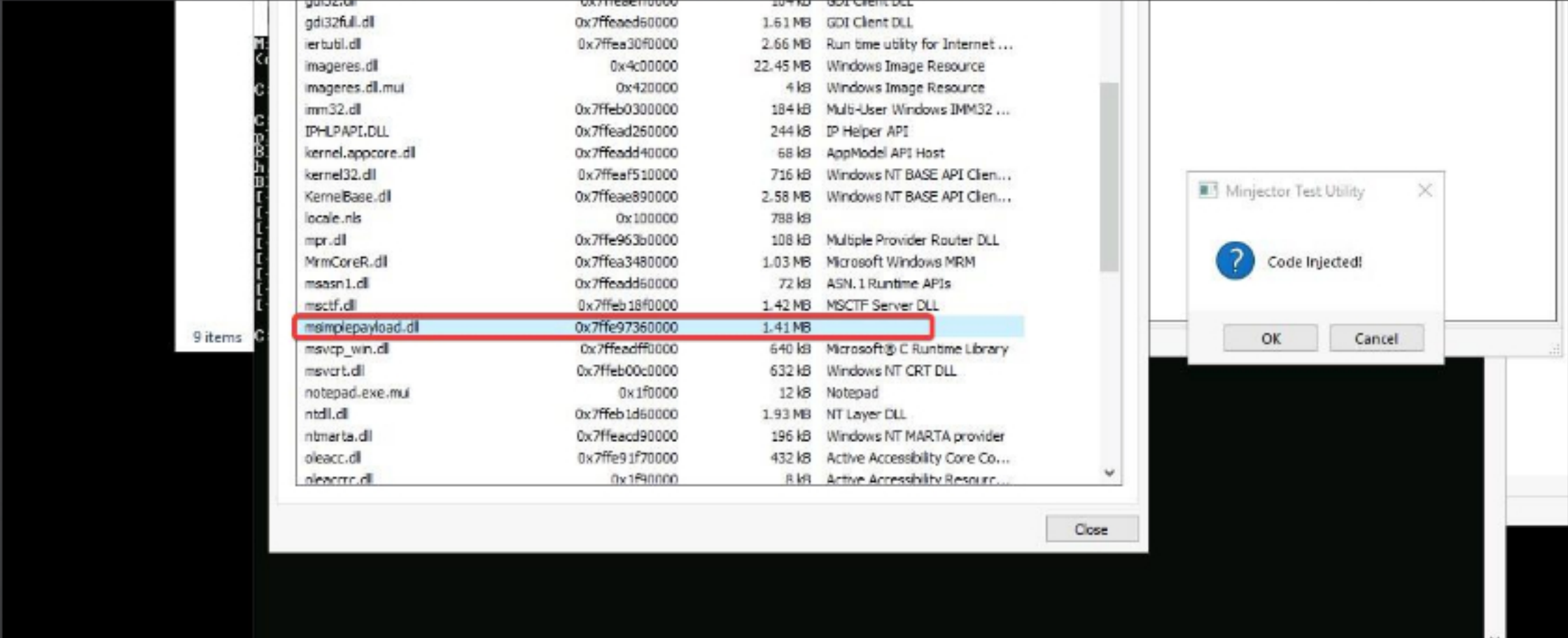
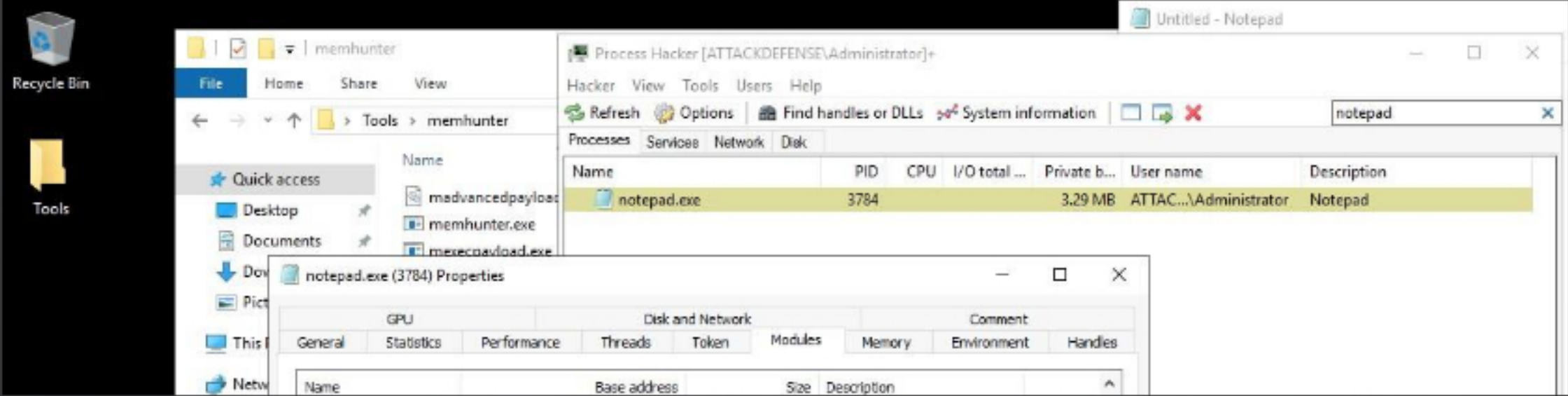
To perform a process injection against notepad.exe, perform the following.

- Run *Process Hacker 2* as an administrator.
- Run notepad.exe and identify its PID through Process Hacker 2
- Open cmd.exe as an administrator and execute:

```
cd C:\Users\Administrator\Desktop\Tools\memhunter
minjector.exe -m 1 -s C:\Users\Administrator\Desktop\Tools\memhunter\msimplepayload.dll -t PID
```



By double-clicking the notepad.exe process inside *Process Hacker 2* and browsing the *"Modules"* tab, we can see the injected dll.

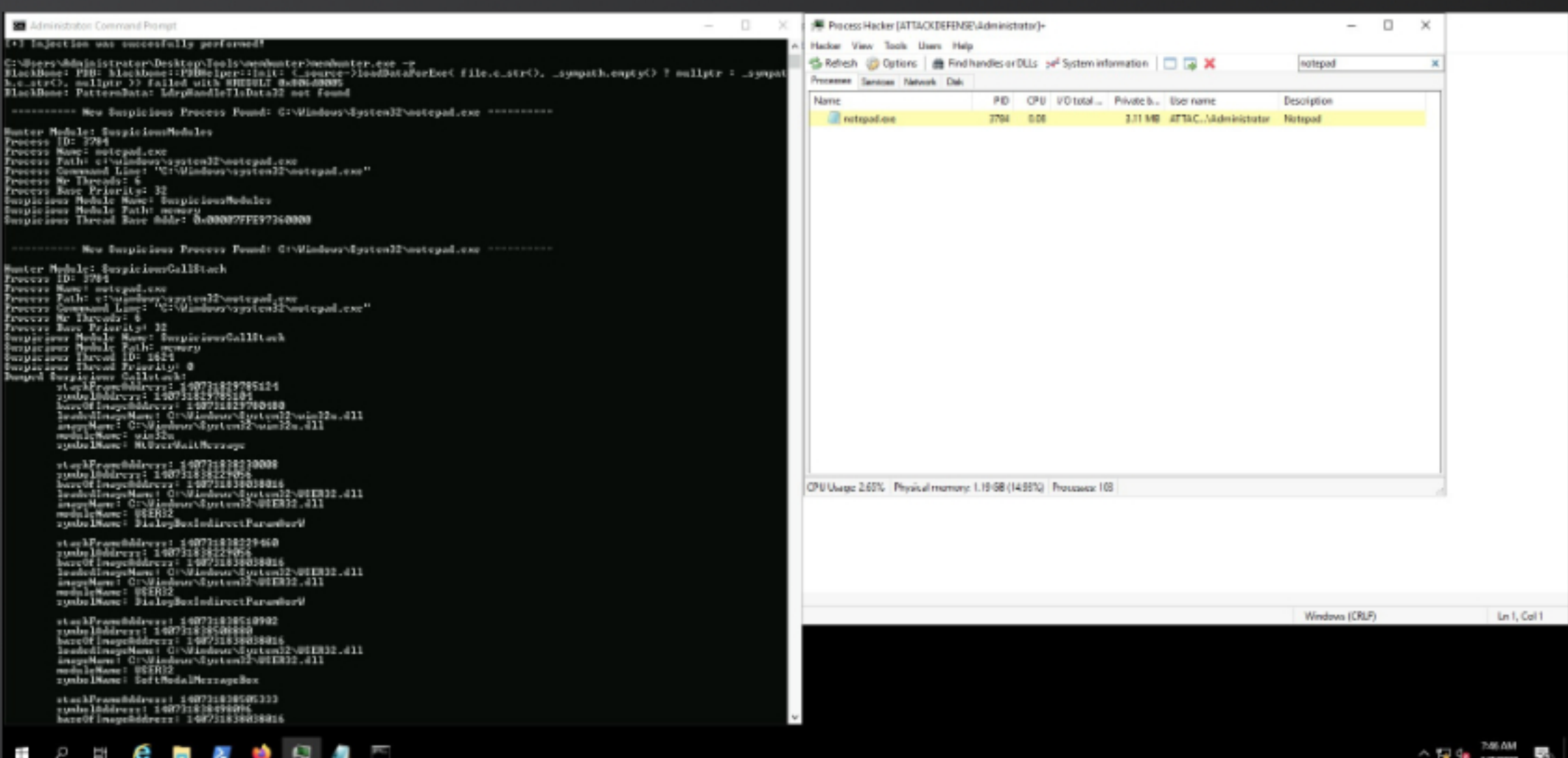


We could have detected the process injection activity through *memhunter.exe*, as follows.

Don't close the notepad.exe process where you injected the DLL previously!!!

Open cmd.exe as an administrator and execute:

```
cd C:\Users\Administrator\Desktop\Tools\memhunter
memhunter.exe -r
```



Wait for 2-3 minutes.

Memhunter successfully identified the process injection activity.

Try to detect process injection through the Reflective DLL injection and Process Hollowing options of minjector.exe with both *Memhunter* and *Process Hacker 2*.

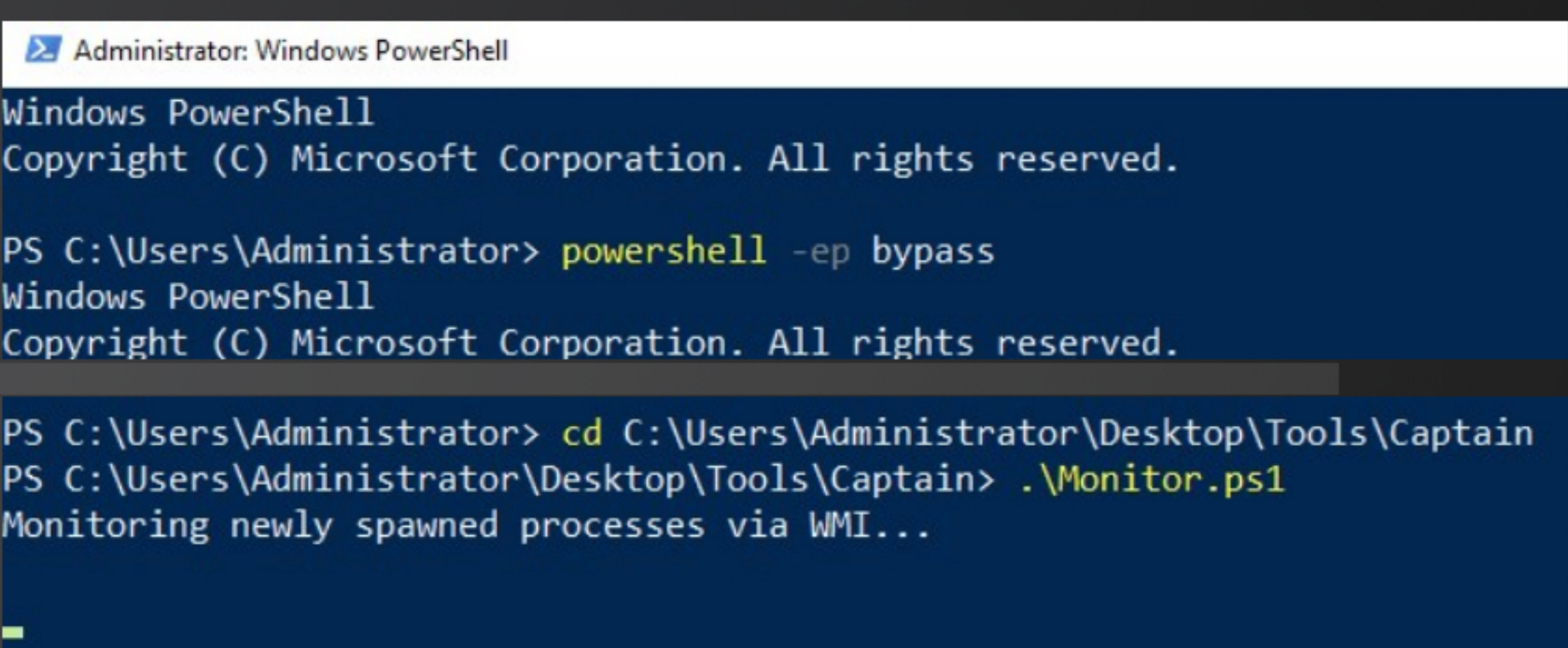
- Regarding Reflective DLL injection, focus on the *"Threads"* tab of *Process Hacker 2*
- Regarding Process Hollowing, focus on the *"General"* and *"Modules"* tabs of *Process Hacker 2*

Task 2. Hunt for suspicious network connections

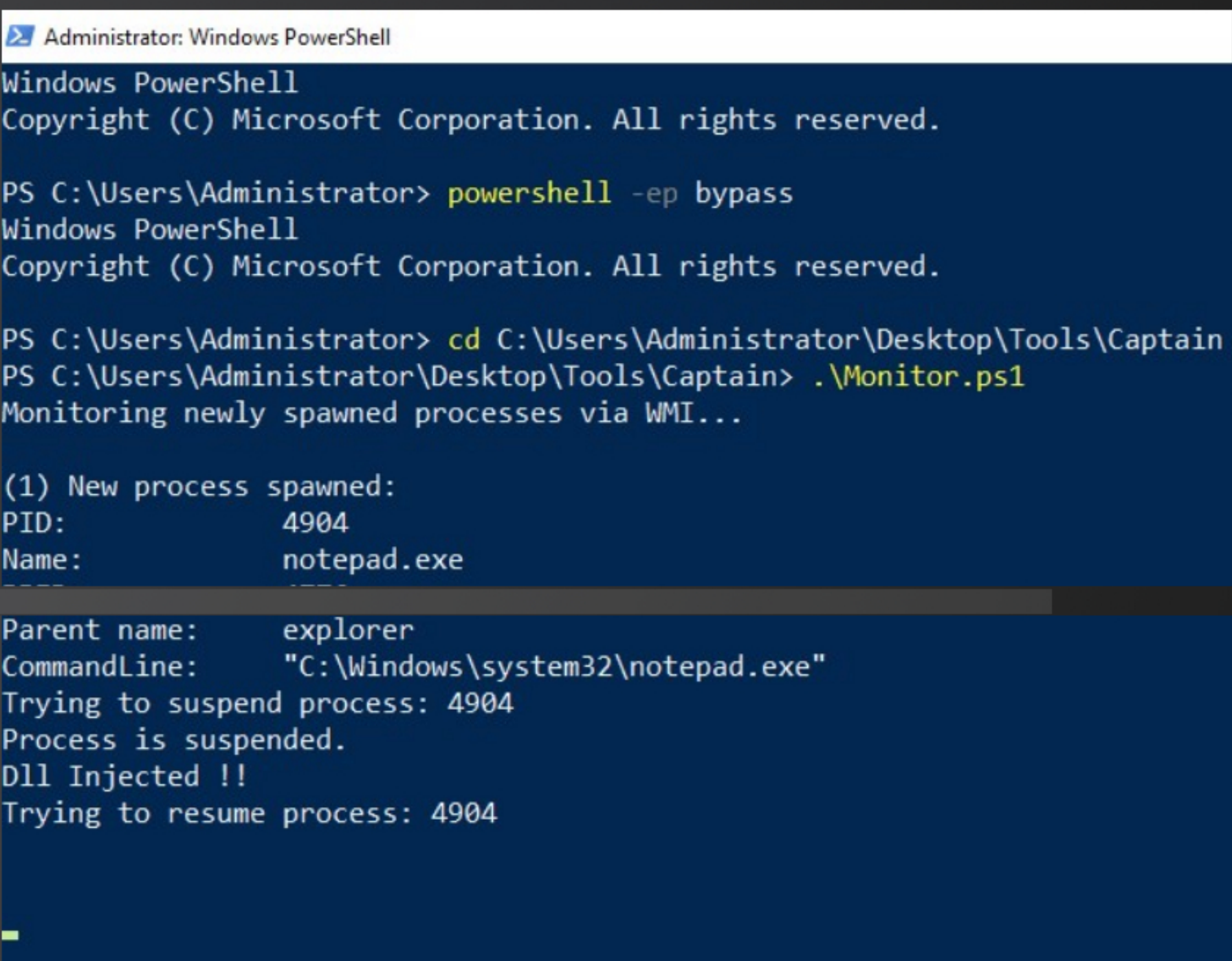
Let's first start Captain's monitoring script, as follows.

Open PowerShell as an administrator, and execute.

```
powershell -ep bypass
cd C:\Users\Administrator\Desktop\Tools\Captain
.\Monitor.ps1
```



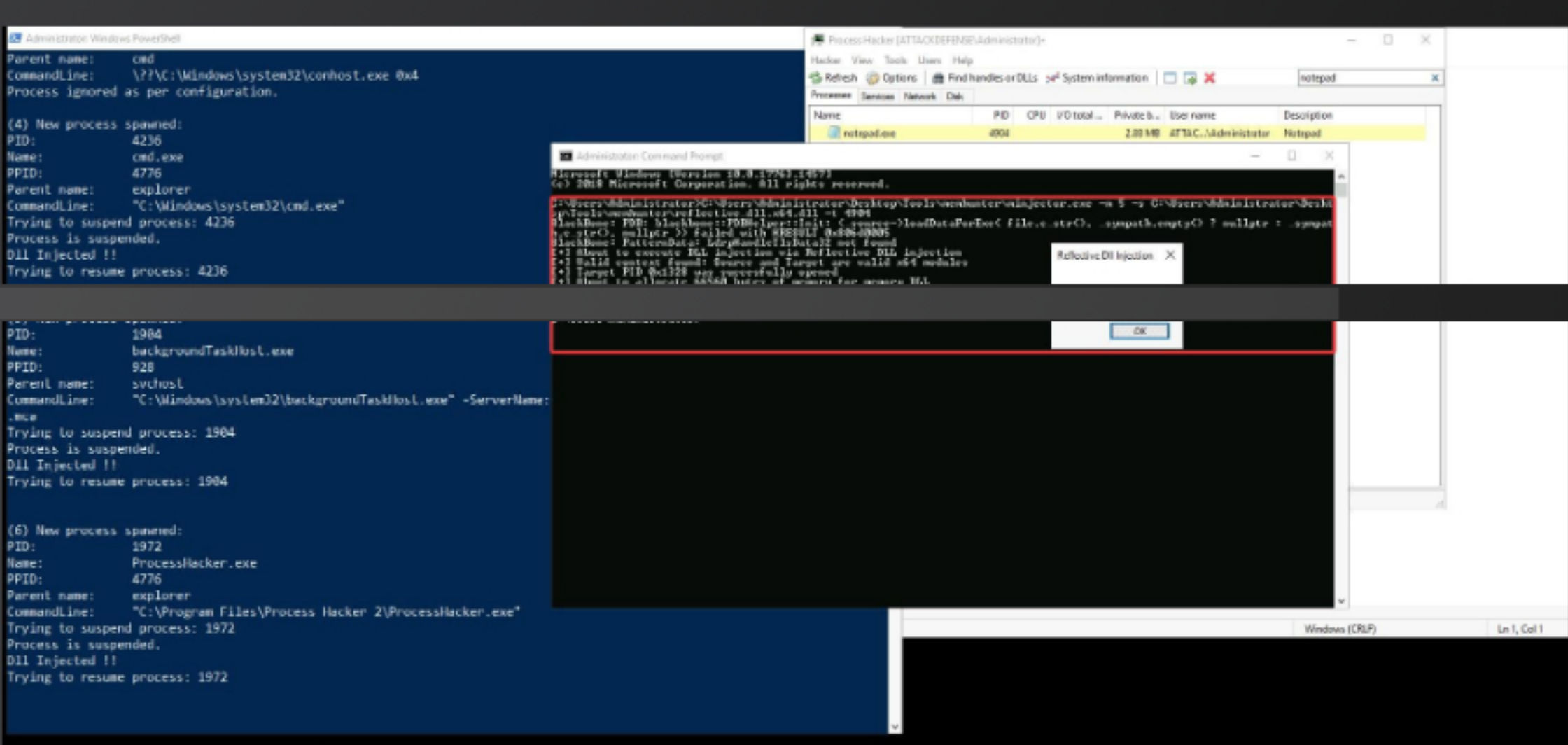
Now, let's run a new notepad.exe process. Once we do that, we should see the below inside Captain's monitoring script.



It is time we perform our Reflective DLL injection against *notepad.exe*, using minjector.exe.

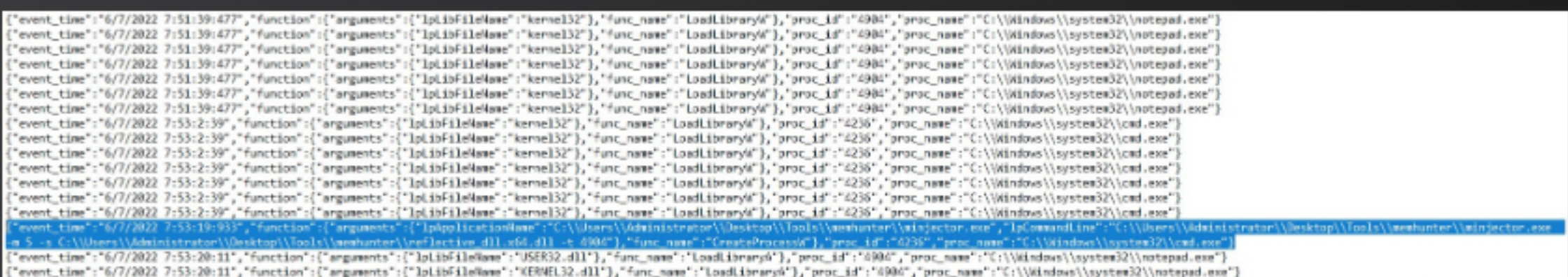
Open cmd.exe as an administrator and execute:

```
cd C:\Users\Administrator\Desktop\Tools\memhunter
minjector.exe -m 5 -s C:\Users\Administrator\Desktop\Tools\memhunter\reflective_dll.x64.dll -t
```



The reflective DLL injection attempt should be successful and a message box mentioning "Hello from DllMain" should be visible (click on the notepad icon on the taskbar if it is not visible).

Finally navigate to *C:\ProgramData\Captain\Reporting* and read the content of *events.json*. You should notice functions that are related with process injection being called and used against *notepad.exe*.



Feel free to explore all functionality of the Captain tool. It will give you a nice idea of how actual EDR work.

<https://github.com/y3n11/Captain>

Not Running

For the best experience, choose the region closest to you. Then, start the lab to begin.

MARK UNFINISHED

Select the region closest to you

US-East

Keyboard layout: USEnglish (US)

Start lab